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BOP2: Bayesian optimal design for phase II clinical trials with simple and complex endpoints

Heng Zhou, J. Jack Lee and Ying Yuan*†

We propose a flexible Bayesian optimal phase II (BOP2) design that is capable of handling simple (e.g., binary) and complicated (e.g., ordinal, nested, and co-primary) endpoints under a unified framework. We use a Dirichlet-multinomial model to accommodate different types of endpoints. At each interim, the go/no-go decision is made by evaluating a set of posterior probabilities of the events of interest, which is optimized to maximize power or minimize the number of patients under the null hypothesis. Unlike other existing Bayesian designs, the BOP2 design explicitly controls the type I error rate, thereby bridging the gap between Bayesian designs and frequentist designs. In addition, the stopping boundary of the BOP2 design can be enumerated prior to the onset of the trial. These features make the BOP2 design accessible to a wide range of users and regulatory agencies and particularly easy to implement in practice. Simulation studies show that the BOP2 design has favorable operating characteristics with higher power and lower risk of incorrectly terminating the trial than some existing Bayesian phase II designs. The software to implement the BOP2 design is freely available at www.trialdesign.org**.** **Copyright © 2017 John Wiley & Sons, Ltd.**

Keywords: Bayesian adaptive design; early stopping; ordinal endpoint; co-primary endpoints; immunotherapy

1. Introduction

The objective of a phase II clinical trial is to evaluate the preliminary efficacy of a new treatment and to determine whether an efficacious treatment warrants investigation in a large-scale randomized phase III trial [1]. A fundamental design feature of phase II clinical trials is the early stopping rule to prevent the exposure of an excessive number of patients to a possibly futile treatment. Numerous designs have been developed for phase II clinical trials. Among the frequentist designs, the most well known is Simon's optimal two-stage design [2], which minimizes the expected sample size or the maximum sample size under the null hypothesis that the treatment is not effective, while controlling the type I and type II error rates at desirable levels. Other related work includes Fleming's multiple-stage test [3], Ensign's optimal three-stage design [4], and Chen's optimal three-stage design [5], among others.

A number of Bayesian designs have been proposed for phase II trials. Thall and Simon [6] proposed using the posterior probability to monitor phase II trials and terminating the trial if the interim data indicate that the response rate for the treatment has high probability of being smaller than a prespecified threshold. That method is simple and intuitive and has been widely used at MD Anderson Cancer Center. On the basis of the posterior probability, Heitjan [7] advocated the use of a 'persuasion probability' as a consistent criterion for determining whether a drug is promising or not. Tan and Machin [8] proposed two Bayesian two-stage designs that mimic frequentist multistage designs. Lee and Liu [9] proposed a Bayesian phase II design based on predictive probability. Cai, Liu, and Yuan [10] proposed a Bayesian phase II trial design that can handle delayed efficacy outcomes through the use of multiple imputation.

Traditionally, phase II clinical oncology trials have focused on binary efficacy endpoints, for example, tumor response, but they have become much more complicated with the advent of novel molecular targeted agents and immunotherapy. The endpoints for such treatments may be ordinal or multivariate, and

Department of Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, Texas 77030, U.S.A.
**Correspondence to: Ying Yuan, Department of Biostatistics, The University of Texas MD Anderson Cancer Center Houston, TX 77030, U.S.A.*
†E-mail: yyuan@mdanderson.org

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the investigators are often interested in simultaneously monitoring multiple types of events in the trial, as illustrated by the following trial examples.

Example 1. *Binary efficacy endpoint* The aim of a phase II trial is to evaluate the efficacy of pembrolizumab in patients with advanced small bowel adenocarcinomas. The primary endpoint is the objective response rate (ORR), defined using Response Evaluation Criteria in Solid Tumors, version 1.1. The treatment is regarded as futile if the ORR is $\leqslant 20\%$ and promising if the ORR is $\geqslant 40\%$. This example is used to illustrate the standard case with a binary efficacy endpoint.

Example 2. *Nested efficacy endpoints* The aim of a phase II clinical trial is to assess the efficacy of nivolumab in patients with Hodgkin's lymphoma who have not experienced a successful outcome following an autologous stem cell transplant. The revised International Working Group Criteria for Malignant Lymphoma [11] is used to define the efficacy of treatments for lymphoma, categorized as one of four levels of decreasing desirability: complete remission (CR), defined as the disappearance of all evidence of disease; partial remission (PR), defined as the regression of measurable disease and no new sites; stable disease (SD), defined as failure to attain CR, PR or progressive disease (PD); and PD, defined as evidence of any new lesion or an increase in lesion volume $\geqslant 50\%$ from the nadir of previously involved sites. In this trial, although both CR and PR are regarded as favorable responses, CR is substantially more desirable. The treatment is regarded as promising if (i) the probability of achieving CR or PR is $> 30\%$ or (ii) the probability of achieving CR is $> 15\%$, where the endpoint of the second condition is a part of the endpoints of the first condition.

Example 3. *Co-primary efficacy endpoints* The primary objective of a phase II trial is to evaluate the efficacy of trebananib administered at 15 mg/kg IV per week in patients with persistent or recurrent carcinoma of the endometrium [12]. The trial has two co-primary efficacy endpoints: the ORR and the event-free survival at 6 months (EFS6). The objective response (OR) is defined using Response Evaluation Criteria in Solid Tumors, version 1.1 [13]. The event-free survival (EFS) is defined as the length of time from the initiation of the treatment to disease progression, death, or beginning a subsequent therapy. The null hypothesis is that the ORR is $\leqslant 10\%$ and EFS6 is $\leqslant 20\%$. In other words, the treatment is regarded as futile only if the ORR is $\leqslant 10\%$ and EFS6 is $\leqslant 20\%$. Clinically significant improvements are defined as a 20% increase in EFS6, or a 15% increase in ORR.

Example 4. *Jointly monitoring efficacy and toxicity* In a phase II clinical trial, patients with recurrent indolent non-follicular lymphoma are treated with lenalidomide in combination with rituximab [14]. Lenalidomide is administered at 20 mg/day for days 1–21, and rituximab is administered at 375 mg/m^2 once on day 14 of every 28 days. The primary efficacy endpoint is the ORR as defined using the 1999 Cheson criteria. Because of large uncertainty regarding the safety of the combination treatment, the trial also monitors dose-limiting toxicity, defined according to the National Cancer Institute Common Terminology Criteria for Adverse Events. The lowest acceptable ORR is 45%, and the highest acceptable toxicity rate is 30%.

We propose a flexible Bayesian optimal phase II (BOP2) design that is capable of handling the aforementioned trials in a unified framework. We use a Dirichlet-multinomial model to embrace different types of endpoints. At each interim, the go/no-go decision is made by evaluating the posterior probabilities of the events of interest. The BOP2 design explicitly controls the type I error rate and is optimal in the sense that it optimizes power under the alternative hypothesis with a given sample size or minimizes the expected sample size under the null hypothesis. Thall and colleagues [15–17] proposed Bayesian sequential monitoring designs for multiple response outcomes (e.g., toxicity and efficacy). Compared with these designs, the advantage of the proposed BOP2 design includes (i) offering a more flexible framework to monitor multiple events simultaneously, including nested or co-primary endpoints; (ii) explicitly controlling the type I error rate, thereby bridging the gap between Bayesian designs and frequentist designs and also rendering the proposed Bayesian design more accessible to a wide range of users and regulatory agencies; and (iii) allowing the cutoffs of the stopping rule to vary with the interim sample size, which improves the power of the design, as demonstrated in our simulation studies.

The remainder of this article is organized as follows. In Section 2, we describe our model and propose the BOP2 design. In Section 3, we describe the graphical user interface-based web application for

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implementing the BOP2 design. In Section 4, we present simulation studies to evaluate the operating characteristics of the BOP2 design and conclude with a brief discussion in Section 5.

2. Methods

2.1. Probability model

Although the endpoints of the aforementioned trials take different forms, they can be unified and represented by a random variable Y that follows a multinomial distribution,

$$Y \sim \text{Multinom}(\theta_1, \dots, \theta_K),$$

where $\theta_k = \Pr(Y = k)$ is the probability that Y belongs to the kth category, $k = 1, \dots, K$. The K categories can be the actual levels of a single endpoint or the combinational levels of multiple categorical endpoints. For example, in trial example 2, Y is the ordinal outcome, with $Y = 1, 2, 3$, and 4 denoting CR, PR, SD, and PD, respectively. In trial example 3, Y is a multinomial variable with four categories where 1 = (OR, EFS6), 2 = (OR, no EFS6), 3 = (no OR, EFS6), and 4 = (no OR, no EFS6). Similarly, in trial example 4, Y is a multinomial variable with four categories: 1 = (toxicity, OR), 2 = (no toxicity, OR), 3 = (toxicity, no OR), and 4 = (no toxicity, no OR). Trial example 1 can be viewed as a special case of trial example 2 by ignoring EFS6, where Y has only two categories (i.e., OR or no OR). In this case, the multinomial distribution degenerates to a binomial distribution.

Suppose that at an interim time, a total of n patients have been enrolled into the trial and their endpoints have been fully evaluated. Let $\mathcal{D}_n = (x_1, \dots, x_K)$ denote the interim data and x_k denote the number of patients with $Y = k$, where $\sum_{k=1}^{K} x_k = n$. Assuming that $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)^T$ follows a Dirichlet prior,

$$(\theta_1, \dots, \theta_K) \sim \text{Dir}(a_1, \dots, a_K),$$

where $a_1, \dots, a_K$ are hyperparameters, the posterior distribution of $\boldsymbol{\theta}$ is given by

$$\boldsymbol{\theta}|\mathcal{D}_n \sim \text{Dir}(a_1 + x_1, \dots, a_K + x_K).$$

We set $\sum_{k=1}^{K} a_k = 1$ such that the prior is vague and equivalent to a prior sample size of 1. In the special case that Y is a binary outcome (e.g., trial example 1), this Dirichlet-multinomial model becomes a standard beta-binomial model.

2.2. Trial design

Let N denote the maximum sample size of the trial. The proposed BOP2 design consists of R interim looks, which occur when the number of enrolled patients reaches $n_1, \dots, n_R$, and a final look when all N patients are enrolled. At each of these looks, the go/no-go decision is made on the basis of the accumulating data, as described in the succeeding texts. In other words, patients are enrolled in $R + 1$ cohorts of size n_1, $n_2 - n_1$, …, $n_R - n_{R-1}$ and $N - n_R$, respectively, and the go/no-go decision is made after each cohort is enrolled. When $R = N - 1$, we obtain a full sequential design in which the go/no-go decision is continuously assessed after each patient. For notational brevity, we suppress the subscript of the interim sample size when this does not cause confusion.

Let $C(n)$ denote a probability cutoff, which is a function of the interim sample size n. Under the proposed design, the go/no-go decision at each interim is made on the basis of the posterior probability of the events of interest. Specifically, for the four trial examples, the interim stopping rule is described as follows. At an interim look, terminate the trial if

(*Example 1*): $\Pr(\theta_1 \leqslant 0.2|\mathcal{D}_n) > C(n)$;
(*Example 2*): $\Pr(\theta_1 \leqslant 0.15|\mathcal{D}_n) > C(n)$ and $\Pr(\theta_1 + \theta_2 \leqslant 0.3|\mathcal{D}_n) > C(n)$;
(*Example 3*): $\Pr(\theta_1 + \theta_2 \leqslant 0.1|\mathcal{D}_n) > C(n)$ and $\Pr(\theta_1 + \theta_3 \leqslant 0.2|\mathcal{D}_n) > C(n)$;
(*Example 4*): $\Pr(\theta_1 + \theta_2 \leqslant 0.45|\mathcal{D}_n) > C(n)$ or $\Pr(\theta_1 + \theta_3 > 0.3|\mathcal{D}_n) > C(n)$.

Unlike most existing Bayesian designs [6, 15, 16], which assume a constant cutoff, here we allow the cutoff $C(n)$ to be a function of the interim sample size n. As we show later, such modification is important and substantially improves the power of the design. Although these stopping rules have different clinical

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Table I. Stopping boundaries of the BOP2 design for four trial examples.

Trial		Stop the trial if	Number of patients treated 10	15	20	25	30	35	40
Example 1		# of ORR ⩽	1	2	4	5	7	9	10
Example 2	and	# of CR ⩽	0	1	3	4	5	7	9
		# of CR/PR ⩽	2	3	5	8	10	13	16
Example 3	and	# of ORR ⩽	0	1	2	3	4	5	7
		# of EFS6 ⩽	1	2	4	5	7	9	12
Example 4	or	# of responses ⩽	2	5	7	10	13	16	19
		# of toxicities ⩾	5	6	8	9	10	11	12

Maximum sample size is 40. Type I error rate is 10%.
BOP2, Bayesian optimal phase II; ORR, objective response rate; CR, complete remission; PR, partial remission; EFS6, event-free survival at 6 months.

interpretations, the go/no-go decisions are all based on the evaluation of a set of the posterior probabilities of the linear combination of the model parameters $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)^T$, for example,

$$\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n) > C(n), \tag{1}$$

where $\boldsymbol{b}$ is a design vector with elements of 0 and 1 and ϕ is a prespecified threshold. For example, in trial example 2, the stopping rule involves the evaluation of two posterior probabilities, with $\boldsymbol{b} = (1, 0, 0, 0)$ and $\phi = 0.15$ and $\boldsymbol{b} = (1, 1, 0, 0)$ and $\phi = 0.3$, respectively, and in trial example 3, the stopping rule involves the evaluation of two posterior probabilities, with $\boldsymbol{b} = (1, 1, 0, 0)$ and $\phi = 0.1$ and $\boldsymbol{b} = (1, 0, 1, 0)$ and $\phi = 0.2$, respectively.

The evaluation of the posterior probability in (1) is facilitated by the following property of the Dirichlet distribution.

Property 1
Given $\theta \sim \mathrm{Dir}(a_1 + x_1, \dots, a_K + x_k)$ and a design vector $\boldsymbol{b} = (b_1, \dots, b_K)$ with elements of 0 and 1, $\boldsymbol{b\theta}$ follows a beta distribution $\mathrm{Beta}\left(\sum_{k=1}^K b_k(a_k + x_k), \sum_{k=1}^K (1 - b_k)(a_k + x_k)\right)$.

As a result, $\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n)$ can be easily evaluated as

$$\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n) = B\left(\phi; \sum_{k=1}^K b_k(a_k + x_k), \sum_{k=1}^K (1 - b_k)(a_k + x_k)\right),$$

where $B(\phi; \zeta, \xi)$ is the cumulative distribution function of a beta distribution with parameters ζ and ξ, evaluated at value ϕ. This property of $\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n)$ leads to the following result.

Lemma 1
$\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n)$ is a monotonic function of $\sum_{k=1}^K b_k x_k$.

The monotonicity of $\Pr(\boldsymbol{b\theta} \leqslant \phi | \mathcal{D}_n)$ is important in practice because it allows us to enumerate the stopping boundary prior to the onset of the trial, similar to Simon's two-stage design, as shown in Table I. For example, the row 'Example 2' shows the interim stopping boundary in terms of the number of patients with CR and CR/PR for our trial example 2. During trial conduct, we do not need to carry out any complicated calculations; rather, we just need to count the number of relevant events and make the go/no-go decision on the basis of whether that count exceeds the boundary. For example, after 20 patients are treated, if the number of CR responses is ⩽ 3 and the number of CR/PR responses is ⩽ 5, we terminate the trial early. This property makes the BOP2 design very easy to implement in practice.

2.3. *Optimizing design parameters*

Suppose that appropriate null hypothesis H_0 and alternative hypothesis H_1 have been chosen to reflect clinical interests, where H_0 specifies the value of $\boldsymbol{\theta}$, under which the treatment is deemed as futile, and H_1 specifies the value of $\boldsymbol{\theta}$, under which the treatment is deemed as promising. For example, for trial

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example 2, $H_0 : \theta_1 = 0.15$ and $\theta_1 + \theta_2 = 0.3$, and a reasonable alternative hypothesis is $H_1 : \theta_1 = 0.25$ and $\theta_1 + \theta_2 = 0.5$. With complicated endpoints (e.g., two co-primary endpoints), the specification of H_1 is less straightforward and should be determined through consultation with clinicians to reflect a desirable outcome that is feasible in practice. We reject H_0 and claim that the treatment is promising if the stopping boundaries are never crossed throughout the trial (including at the end of the trial). The type I error rate and statistical power are defined as the probability of rejecting H_0 under H_0 and H_1, respectively.

The operating characteristics of the BOP2 design rely on the specification of the probability cutoff $C(n)$. Although any reasonably flexible monotonically decreasing function may be used, one particular function of $C(n)$ that is simple and yields good operating characteristics is the following two-parameter power function:

$$C(n) = 1 - \lambda(n/N)^{\gamma}, \tag{2}$$

where λ and γ are tuning parameters. We require that $\gamma > 0$ such that $C(n)$ is monotonically decreasing with n/N, the fraction of the accumulated information. The rationale is that at the beginning of the trial, data are sparse and a more relaxed stopping rule with a larger value of $C(n)$ may be preferred to avoid terminating the trial accidentally. When the trial proceeds and information accumulates, we have less uncertainty regarding the endpoint of interest, and thus, it is desirable to have a more stringent stopping rule with a smaller value of $C(n)$ to terminate the trial for an inefficacious treatment. The remaining questions are how to choose the tuning parameters λ and γ and sometimes N in (2) to optimize the performance of the design according to a certain criterion.

We first consider how to choose the tuning parameters λ and γ when sample size N is fixed, for example, because of a fixed budget or limited accrual. Our strategy is to choose λ and γ to maximize the power of the BOP2 design while controlling the type I error rate at a certain prespecified level. This can be carried out as follows:

Step 1: Elicit from clinicians H_0 and H_1 and the desirable type I error rate.
Step 2: Find the values of (λ, γ) that yield the desirable type I error rate, which can be carried out through a grid search.
Step 3: Among the set of (λ, γ) identified in step 2, select the one that yields the maximum statistical power as the optimal design parameters.

Although the BOP2 design is a Bayesian design, it is still essential to ensure that the design has desirable frequentist operating characteristics (e.g., type I rate and power). In general, a good Bayesian design should demonstrate reasonable frequentist operating characteristics [18]. Explicitly controlling the type I error rate is an important feature that distinguishes the BOP2 design from most existing Bayesian phase II designs. This feature bridges the gap between Bayesian designs and frequentist designs, making the BOP design accessible to a wide range of users and regulatory agencies.

An alternative optimization strategy is to choose λ and γ, as well as sample size N, to minimize the expected sample size under H_0, that is, $E(N|H_0)$, given prespecified types I and II error rates. This optimization criteria was used in Simon's optimal design. In this approach, N is not fixed, but a design parameter is to be optimized. The procedure to determine the values of (λ, γ, N) that minimize $E(N|H_0)$ can be described as follows:

Step 1: Elicit from clinicians H_0 and H_1 and desirable types I and II error rates.
Step 2: Find the values of (N, λ, γ) that yield the desirable types I and II error rates, which can be carried out through a grid search.
Step 3: Among the set of (N, λ, γ) identified in step 2, select the one that yields the smallest $E(N|H_0)$ as the optimal design parameters.

In step 2, we have two constraints (i.e., types I and II error rates), but need to determine the values of three unknown parameters (N, λ, γ). Thus, in principle, there are an infinite number of possible solutions. We circumvent this issue by restricting the value of N within the range of $(N_{\min}, N_{\max})$, where $N_{\max}$ is the maximum sample size that we can afford in practice, which is often determined by budget, accrual rate, or other practical factors. $N_{\min}$ is the minimal sample size for the trial, which has little impact on the operating characteristics of the design as long as it is reasonably small, such as $N_{\min} = 10$. Given a specific value of N, we can uniquely determine the values of λ and γ based on the two constraints through a grid search. One potential limitation of this optimization strategy is that we do not have a direct control on sample size N, and the value of N that minimizes $E(N|H_0)$ may be excessively large for practical use

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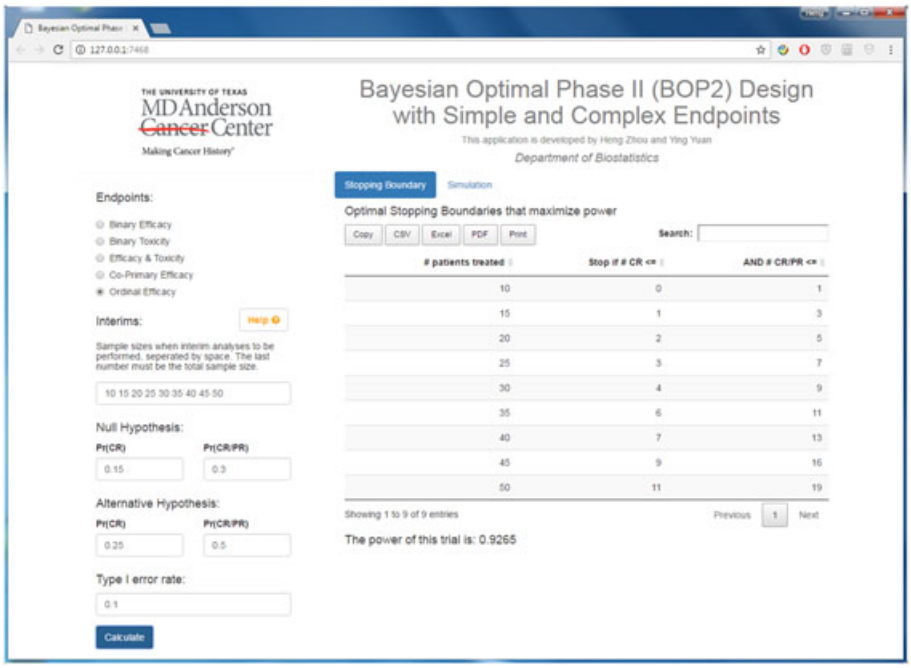

Figure 1. Web application for the BOP2 design. [Colour figure can be viewed at wileyonlinelibrary.com]

in some cases. When this is a concern, the minimax criterion can be used to optimize the design. That is, instead of minimizing $E(N|H_0)$, we choose (λ, γ, N) to minimize the maximum sample size.

3. Web application

To facilitate the use of the BOP2 design, we develop an easy-to-use web application using Shiny. Figure 1 shows the graphical user interface of the application. After users input their design parameters (e.g., the maximum sample size, cohort size, desirable types I and II errors, and the type of endpoints to be monitored), the web application generates the operating characteristics and stopping boundary of the BOP2 design that can be included in the trial protocol. As described previously, similar to Simon's two-stage design, one important advantage of the BOP2 design is that its stopping boundary can be enumerated and included in the trial protocol prior to the onset of the trial. When conducting the trial, we simply count the number of relevant events and make the go/no-go decision on the basis of whether that count exceeds the boundary. Our web application will be freely available at http://www.trialdesign.org and also at the author's website.

4. Simulation studies

In the following simulation studies, we controlled the type I error rate at 0.1 for all designs. Because of the discrete nature of the observed data, a cutoff that yields a type I error rate of exactly 0.1 does not usually exist. In these cases, we chose the cutoff that yielded a type I error rate closest to and not higher than the nominal value. For clarity, in what follows, we focus on the BOP2 design that maximizes power. Interim analyses were carried out after the first 10 patients were treated, then after every 5 additional patients were treated, with the maximum sample size $N = 40$. The results of the BOP2 design that minimizes $E(N|H_0)$ are similar and provided in the Appendix.

To evaluate the performance of the designs, we considered the following three metrics. (i) The percentage of rejecting the null (PRN) hypothesis is defined as the percentage of the simulated trials in which H_0 is rejected. The PRN is the type I error rate (or power) when H_0 is (or is not) true. The PRN can be also explained as the percentage of claiming that the new treatment is effective. (ii) The percentage of early termination (PET) is defined as the percentage of trials that are terminated early. (iii) The actual sample size is defined as the average sample size actually used in 10,000 simulated trials.

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4.1. Binary efficacy endpoint

We first evaluated the operating characteristics of the BOP2 design under the conventional setting with a simple binary efficacy outcome (i.e., OR/no OR), as illustrated by trial example 1. We selected four pairs of H_0 and H_1 and compared the BOP2 design with the Bayesian design proposed by Thall and Simon [6], denoted as the TS design. In the first scenario, for example, the TS design employed a beta-binomial model and terminated the trial if $\Pr(\theta_1 \leqslant 0.2|\mathcal{D}_n) > C$, where C is a fixed cutoff. We calibrated the value of C to control the type I error rate of the TS design at 0.1, which matches that of the BOP2 design. Table I provides the stopping boundaries of the BOP2 design in terms of the observed number of patients who had responses, which were used to make the go/no-go decision at each interim look for scenario 1 in Table II.

Table II shows the performance of the two designs under four different pairs of H_0 and H_1 on the ORR, represented as scenarios 1–4. In general, the BOP2 design yielded substantially higher power than the TS design. For example, in scenario 1, where the null ORR is 0.2 and the alternative ORR is 0.4, when the true ORR is 0.4, the power of the BOP2 design is 88.3%, whereas the power of the TS design is only 76.4%. In addition, compared with the TS design, the BOP2 design had a lower risk of incorrectly terminating the trial when the treatment is actually effective. For example, when the true ORR is 0.4, the TS design incorrectly terminated the trial 23.5% of the time, while the BOP2 design incorrectly terminated the trial 11.4% of the time. Under the null hypothesis, the TS design had a higher probability of terminating the trial than the BOP2 design. Because the TS design had a high tendency of terminating the trial, it has smaller (actual) sample sizes than the BOP2 design. Because of the discrete nature of the observed data and different ways of defining the cutoff (i.e., the TS design uses a fixed cutoff C and the BOP2 design uses an adaptive cutoff $C(n)$), in some cases, it is not possible to exactly match the type I error rate of two designs to 0.1. That is why in some scenarios (e.g., scenario 1), the type I error of the TS design is slightly lower than the BOP2 design but slightly higher in other scenarios (e.g., scenario 2).

Figure 2 contrasts the stopping boundaries of the two designs under scenario 1 in Table II. At each interim look, the trial is terminated if the number of observed responses does not exceed the line in the plot. In the early stages of the trial, the BOP2 design has lower boundaries because of the relaxed stopping rules. At later stages, the BOP2 design applies more stringent stopping rules to yield higher boundaries than the TS design. Similar patterns are observed in the boundary plots for other scenarios in Table II.

Table II. PRN, PET, and actual sample size under the BOP2 design and TS design (Thall and Simon, 1994) with a binary endpoint as described in trial example 1.

	Response rate	PRN (%)		PET (%)		Sample size	
Scenario	(ORR)	BOP2	TS	BOP2	TS	BOP2	TS
1	0.20[§]	9.6	9.4	88.8	89.8	20.2	15.3
	0.30	55.2	42.6	46.2	56.7	31.0	24.9
	0.40[†]	88.3	76.4	11.4	23.5	37.6	33.6
	0.50	98.2	93.3	1.8	6.7	39.5	38.1
2	0.30[§]	9.4	10.0	82.8	89.5	22.2	15.6
	0.40	48.2	40.3	41.4	59.4	31.9	24.3
	0.50[†]	86.7	74.0	10.3	25.9	37.7	32.9
	0.60	98.9	92.7	1.8	7.3	39.5	37.9
3	0.40[§]	10.0	10.0	84.3	89.0	21.6	15.4
	0.50	47.5	38.3	46.0	61.1	31.0	23.5
	0.60[†]	86.3	72.1	11.9	27.9	37.5	32.4
	0.70	98.2	92.5	1.7	7.5	39.6	37.9
4	0.50[§]	7.2	7.2	79.5	92.4	24.7	13.4
	0.60	43.0	29.6	36.3	70.2	33.9	20.0
	0.70[†]	87.3	61.9	6.4	38.1	38.9	28.9
	0.80	99.6	87.6	0.2	12.4	40.0	36.3

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; ORR, objective response rate.

[§] null hypothesis.

[†] alternative hypothesis.

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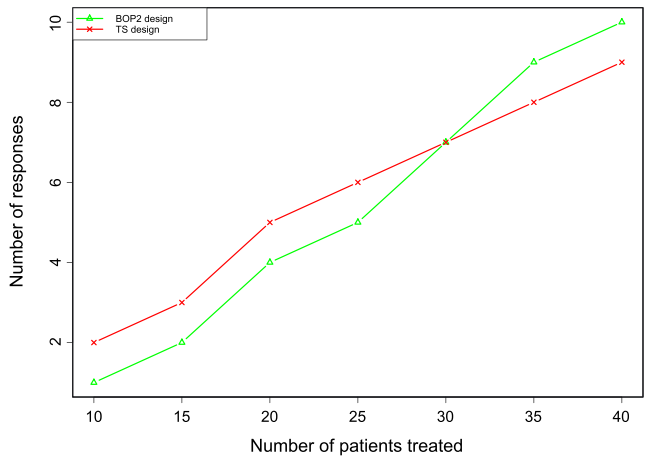

Figure 2. Stopping boundaries of the Bayesian optimal phase II (BOP2) design and Thall and Simon (TS) design for the binary efficacy endpoint (i.e., trial example 1) under scenario 1 shown in Table II. The maximum sample sizes of the two designs are 40. [Colour figure can be viewed at wileyonlinelibrary.com]

Table III. PRN, PET, and the actual sample size under the BOP2 design and TS design (Thall and Simon, 1994) with nested endpoints as described in trial example 2.

			PRN (%)		PET (%)		Sample size	
Scenario	$\boldsymbol{\theta}$	(CR, CR/PR)	BOP2	TS	BOP2	TS	BOP2	TS
1	(0.15, 0.15, 0.30, 0.40)[§]	(0.15, 0.30)	8.7	9.9	82.1	89.6	25.4	15.7
2	(0.20, 0.10, 0.30, 0.40)	(0.20, 0.30)	24.2	9.6	63.8	89.9	29.0	15.6
3	(0.20, 0.15, 0.30, 0.35)	(0.20, 0.35)	30.6	22.9	56.3	76.6	31.0	19.7
4	(0.20, 0.20, 0.30, 0.30)	(0.20, 0.40)	45.9	40.7	41.0	59.2	33.8	24.4
5	(0.20, 0.25, 0.30, 0.25)	(0.20, 0.45)	66.3	58.7	24.1	41.2	36.4	29.0
6	(0.25, 0.20, 0.30, 0.25)	(0.25, 0.45)	72.3	59.0	19.3	40.9	37.1	29.1
7	(0.25, 0.25, 0.25, 0.25)[†]	(0.25, 0.50)	85.5	74.2	9.9	25.7	38.5	33.0
8	(0.30, 0.25, 0.25, 0.20)	(0.30, 0.55)	95.7	85.2	3.0	14.8	39.5	35.9

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; CR, complete remission; PR, partial remission.

[§] null hypothesis.

[†] alternative hypothesis.

4.2. *Nested efficacy endpoints*

Table III shows the simulation results for the ordinal endpoint under the setting of trial example 2, and Table I shows the corresponding boundaries. Interim monitoring started after the first 10 patients were enrolled and then was performed after every 5 patients was enrolled. The null hypothesis is scenario 1, that is, H_0: $\Pr(CR) = 0.15$ and $\Pr(CR/PR) = 0.3$, and the alternative hypothesis is scenario 7, that is, H_1: $\Pr(CR) = 0.25$ and $\Pr(CR/PR) = 0.5$. We compared the BOP2 design with the TS design, which regards CR/PR as response and SD/PD as nonresponse, as is often carried out in practice. Thus, the TS design employed the beta-binomial model and terminated the trial if $\Pr(\theta_1 \leqslant 0.3|\mathcal{D}_n) > C$, where C is a fixed cutoff. We calibrated the value of C to control the type I error rate of the TS design at 0.1.

As shown in Table III, the BOP2 design generally has more power than the TS design. For example, in scenario 7, where $\Pr(CR) = 0.25$ and $\Pr(CR/PR) = 0.5$, the power of the BOP2 design is 85.5%, whereas that of the TS design is 74.2%. Comparing the first two scenarios, we observe that the BOP2 design can increase the PRN from 8.7% to 24.2% because the true CR rate increases from 0.15 to 0.20. This exactly fulfills our expectation of the BOP2 design, which can monitor nested endpoints simultaneously. In contrast, the TS design could not distinguish these two scenarios because the CR/PR rates are both 0.30. In addition, the TS design tended to incorrectly terminate the trial more frequently than the BOP2 design when the treatment actually was effective. For example, in scenario 7, the TS design terminated the trial early 25.7% of the time, whereas the BOP2 design terminated the trial early 9.9% of the time. Again, because of a high tendency of terminating the trial early, the actual sample size of the TS design is smaller than that of the BOP2 design.

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4.3. Co-primary efficacy endpoints

Table IV shows the simulation results under the setting of trial example 3, with two co-primary efficacy endpoints (i.e., ORR and EFS6), and Table I shows the corresponding boundaries. The H_0 and H_1 are scenarios 1 and 7, respectively. We compared the BOP2 design with a Bayesian design inspired by the method of Thall, Simon, and Estey [15], and denoted the latter as the TSE design. For fair comparison, the TSE design used the same model and stopping rule as the BOP2 design, except that a fixed cutoff C was used in the stopping rule, as suggested by Thall, Simon, and Estey [15]. The results are generally similar to those described previously. That is, the BOP2 design yielded higher power and was less likely to incorrectly terminate the trial than the TSE design.

4.4. Efficacy and toxicity endpoints

Table V shows the simulation results under the setting of trial example 4, where we simultaneously monitored efficacy and toxicity. Scenarios 1 and 7 show H_0 and H_1, respectively. We compared the BOP2 design with the TSE design, which used the same model and stopping rule as the BOP2 design, except that a fixed cutoff C was used in the stopping rule. Again, given the same type I error rate (i.e., PRN in scenario 1), the BOP2 design outperformed the fixed cutoff design with higher power and smaller risk of incorrectly terminating the trial.

The corresponding boundaries are shown in Table I. The use of stopping boundaries in this case is slightly different from that in the previous cases. For example, after 30 patients are treated, the BOP2 design terminates the trial if either the number of responses is $\leqslant 13$ or the number of toxicities is $\geqslant 10$.

Table IV. PRN, PET, and the actual sample size under the BOP2 design and TSE design (Thall, Simon and Estey, 1995) with two co-primary efficacy endpoints as described in trial example 3.

			PRN (%)		PET (%)		Sample size	
Scenario	$\boldsymbol{\theta}$	(ORR, EFS6)	BOP2	TSE	BOP2	TSE	BOP2	TSE
1	(0.05, 0.05, 0.15, 0.75)[§]	(0.10, 0.20)	7.2	7.3	80.9	92.3	24.5	13.7
2	(0.05, 0.10, 0.15, 0.70)	(0.15, 0.20)	23.9	17.4	58.9	82.3	29.7	16.5
3	(0.10, 0.10, 0.15, 0.65)	(0.20, 0.25)	56.5	37.1	29.7	62.8	35.0	22.0
4	(0.10, 0.15, 0.15, 0.60)	(0.25, 0.25)	79.7	53.0	12.8	47.0	37.8	26.3
5	(0.10, 0.15, 0.20, 0.55)	(0.25, 0.30)	85.9	60.7	7.6	39.3	38.7	28.5
6	(0.15, 0.15, 0.10, 0.60)	(0.30, 0.25)	91.3	64.4	6.6	35.6	38.6	29.5
7	(0.15, 0.15, 0.20, 0.50)[†]	(0.30, 0.35)	96.1	75.5	2.4	24.5	39.5	32.8
8	(0.15, 0.15, 0.25, 0.45)	(0.30, 0.40)	98.5	82.6	0.8	17.4	39.8	34.8

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; ORR, objective response rate; EFS6, event-free survival at 6 month.
[§] null hypothesis.
[†] alternative hypothesis.

Table V. PRN, PET, and the actual sample size under the BOP2 design and the TSE design (Thall, Simon, and Estey, 1995) with jointly monitoring efficacy and toxicity endpoints as described in trial example 4.

			PRN (%)		PET (%)		Sample size	
Scenario	$\boldsymbol{\theta}$	(Eff, Tox)	BOP2	TSE	BOP2	TSE	BOP2	TSE
1	(0.15, 0.30, 0.15, 0.40)[§]	(0.45, 0.30)	9.3	9.2	85.3	89.6	22.1	15.2
2	(0.20, 0.30, 0.15, 0.35)	(0.50, 0.35)	7.5	7.1	88.5	91.3	20.6	14.6
3	(0.10, 0.30, 0.15, 0.45)	(0.40, 0.25)	6.7	7.5	88.7	91.4	21.2	14.7
4	(0.15, 0.35, 0.10, 0.40)	(0.50, 0.25)	30.7	25.5	61.7	73.3	28.2	20.1
5	(0.15, 0.35, 0.05, 0.45)	(0.50, 0.20)	41.0	33.8	51.2	65.3	30.7	22.4
6	(0.15, 0.40, 0.05, 0.40)	(0.55, 0.20)	60.8	48.6	33.5	50.8	33.9	26.1
7	(0.18, 0.42, 0.02, 0.38)[†]	(0.60, 0.20)	74.6	59.8	22.0	39.7	36.0	29.1
8	(0.15, 0.50, 0.05, 0.30)	(0.65, 0.20)	82.6	69.4	15.1	30.3	37.0	31.7

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; Eff, efficacy; Tox, toxicity.
[§] null hypothesis.
[†] alternative hypothesis.

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5. Discussion

We have proposed a flexible Bayesian optimal design for phase II trials with simple and complex endpoints under a unified framework. Our BOP2 design explicitly controls the types I and II error rates, thereby bridging the gap between Bayesian designs and frequentist designs. In addition, unlike many existing Bayesian designs, the BOP2 design allows the stopping rule to vary with the interim sample size, which improves the power of the design and reduces the risk of incorrectly terminating the trial when the treatment is actually effective. Another important advantage of the BOP2 design is that the stopping boundary can be enumerated a priori before the onset of the trial, making the BOP2 design particularly easy to implement in practice.

The BOP2 design is flexible in terms of the number of interim looks to be conducted during the trial. The choice of the number of interim looks hinges on logistical considerations. For example, if the endpoints are quickly ascertainable and frequent updating of the interim data is feasible, full sequential or frequent interim monitoring may be used. However, if the endpoints take a long time to be evaluated, we may limit the number of interim looks to avoid frequently suspending the accrual of patients. When desirable, the method of Cai *et al*. [10] can be adopted to eliminate the requirement of suspending the accrual. The BOP2 design focuses on categorical endpoints. Extending it to time-to-event endpoints warrants further research. The BOP2 design is optimized by assuming a parametric form for the posterior probability cutoff $C(n)$.

Another useful extension of the proposed design to allow arbitrary $C(n)$ and develop fully optimal stopping rules along the lines of the works of Eales and Jennison [19] and Jennison and Turnbull [20]. An interesting implication of allowing $C(n)$ to have full flexibility is that the prior distribution plays no part in the quantity to be optimized. This is because according to Lemma 1, monitoring of $\Pr(\boldsymbol{b\theta} \leqslant \phi|\mathcal{D}_n)$ is equivalent to monitoring the sum of the appropriately weighted responses, so that the design using arbitrary $C(n)$ compares these with an arbitrary boundary to maximize power or minimize $E(n|H_0)$ subject to type I or/and type II error requirements.

Appendix A: Simulation result when BOP2 minimizes $E(N|H_0)$ with binary efficacy endpoint

Table VI. PRN, PET, and actual sample size under the BOP2 design and TS design (Thall and Simon, 1994) with a binary endpoint as described in trial example 1.

	Response rate	PRN (%)		PET (%)		Sample size	
Scenario	(ORR)	BOP2	TS	BOP2	TS	BOP2	TS
1	0.20[§]	9.4	9.0	87.5	91.0	21.5	14.2
	0.30	55.1	40.7	40.2	59.3	31.5	22.6
	0.40[†]	90.5	73.0	8.4	27.0	36.6	30.9
	0.50	98.8	91.0	1.1	9.0	37.8	35.6
2	0.30[§]	9.4	9.5	89.1	90.5	23.9	16.2
	0.40	53.8	40.8	44.1	59.2	38.8	28.1
	0.50[†]	90.3	73.3	9.5	26.7	47.2	39.8
	0.60	98.9	91.5	1.1	8.5	49.6	46.7
3	0.40[§]	10.0	9.4	87.5	90.6	24.5	15.2
	0.50	51.7	37.8	44.2	62.2	35.9	24.4
	0.60[†]	89.8	71.1	9.1	28.9	42.3	34.7
	0.70	99.4	91.8	0.6	8.2	43.8	41.3
4	0.50[§]	9.5	9.8	88.7	90.2	23.3	15.4
	0.60	47.6	38.8	49.1	61.2	34.2	24.4
	0.70[†]	90.1	73.9	8.8	26.1	41.4	34.9
	0.80	99.5	94.1	0.5	5.9	42.9	41.1

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; ORR, objective response rate.

[§] null hypothesis.

[†] alternative hypothesis.

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Table AI. Stopping boundaries of the BOP2 design % and TS design (Thall and Simon, 1994) with binary endpoints as described in trial example 1.

# of patients treated	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Stop the trial if # of ORR ⩽	0	1	1	1	1	2	2	2	2	3	3	3	4	4	4
# of patients treated	25	26	27	28	29	30	31	32	33	34	35	36	37	38	
Stop the trial if # of ORR ⩽	4	5	5	5	6	6	7	7	7	8	8	9	9	10	

Maximum sample size is 38, and cohort size is 1.
BOP2, Bayesian optimal phase II; ORR, objective response rate.

Appendix B: Simulation result when BOP2 minimizes $E(N|H_0)$ with nested efficacy endpoint

Table BI. PRN, PET, and the actual sample size under the BOP2 design and TS design (Thall and Simon, 1994) with nested endpoints as described in trial example 2.

			PRN (%)		PET (%)		Sample size	
Scenario	θ	(CR, CR/PR)	BOP2	TS	BOP2	TS	BOP2	TS
1	(0.15, 0.15, 0.30, 0.40)[§]	(0.15, 0.30)	9.3	10.0	83.6	89.0	27.4	17.2
2	(0.20, 0.10, 0.30, 0.40)	(0.20, 0.30)	23.9	10.2	66.8	88.8	32.0	17.2
3	(0.20, 0.15, 0.30, 0.35)	(0.20, 0.35)	32.7	25.3	56.5	73.8	35.8	23.2
4	(0.20, 0.20, 0.30, 0.30)	(0.20, 0.40)	52.3	44.0	38.8	55.6	40.4	29.8
5	(0.20, 0.25, 0.30, 0.25)	(0.20, 0.45)	74.3	62.6	21.1	37.2	44.5	36.1
6	(0.25, 0.20, 0.30, 0.25)	(0.25, 0.45)	78.2	61.5	17.3	38.3	45.4	35.8
7	(0.25, 0.25, 0.25, 0.25)[†]	(0.25, 0.50)	90.0	75.8	8.3	24.2	47.6	40.8
8	(0.30, 0.25, 0.25, 0.20)	(0.30, 0.55)	96.6	86.4	3.0	13.6	49.0	44.8

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; CR, complete remission; PR, partial remission.
[§] null hypothesis.
[†] alternative hypothesis.

Table BII. Stopping boundaries of the BOP2 design with nested endpoints as described in trial example 2.

		Number of patients treated								
	Stop the trial if	10	15	20	25	30	35	40	45	50
	# of CR ⩽	1	1	2	4	5	6	7	9	11
and	# of CR/PR ⩽	2	3	5	7	9	11	14	16	19

Maximum sample size is 50, and cohort size is 5.
BOP2, Bayesian optimal phase II; CR, complete remission; PR, partial remission.

Appendix C: Simulation result when BOP2 minimizes $E(N|H_0)$ with co-primary efficacy endpoint

Table CI. PRN, PET, and the actual sample size under the BOP2 design and TSE design (Thall, Simon and Estey, 1995) with two co-primary efficacy endpoints as described in trial example 3.

			PRN (%)		PET (%)		Sample size	
Scenario	θ	(ORR, EFS6)	BOP2	TSE	BOP2	TSE	BOP2	TSE
1	(0.05, 0.05, 0.15, 0.75)[§]	(0.10, 0.20)	9.1	9.5	76.7	86.8	17.0	13.0
2	(0.05, 0.10, 0.15, 0.70)	(0.15, 0.20)	23.9	19.7	61.2	77.2	20.0	14.9
3	(0.10, 0.10, 0.15, 0.65)	(0.20, 0.25)	50.3	38.8	38.2	59.5	23.7	18.4
4	(0.10, 0.15, 0.15, 0.60)	(0.25, 0.25)	69.7	54.5	23.8	44.7	26.0	21.2
5	(0.10, 0.15, 0.20, 0.55)	(0.25, 0.30)	77.1	62.3	17.2	37.1	27.1	22.7
6	(0.15, 0.15, 0.10, 0.60)	(0.30, 0.25)	81.8	65.2	16.0	34.6	27.3	23.2
7	(0.15, 0.15, 0.20, 0.50)[†]	(0.30, 0.35)	90.1	76.3	8.2	23.6	28.6	25.3
8	(0.15, 0.15, 0.25, 0.45)	(0.30, 0.40)	94.3	82.3	4.7	17.7	29.2	26.5

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; ORR, objective response rate; EFS6, event-free survival at 6 month.
[§] null hypothesis.
[†] alternative hypothesis.

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Table CII. Stopping boundaries of the BOP2 design with two co-primary efficacy endpoints as described in trial example 3.

		Number of patients treated		
	Stop the trial if	10	20	30
and	# of ORR ⩽	1	3	5
	# of EFS ⩽	2	5	9

Maximum sample size is 30, and cohort size is 10.
BOP2, Bayesian optimal phase II; ORR, objective response rate; EFS, event-free survival.

Appendix D: Simulation result when BOP2 minimizes $E(N|H_0)$ with efficacy and toxicity endpoint

Table DI. PRN, PET, and the actual sample size under the BOP2 design and the TSE design (Thall, Simon and Estey, 1995) with jointly monitoring efficacy and toxicity endpoints as described in trial example 4.

			PRN (%)		PET (%)		Sample size	
Scenario	$\boldsymbol{\theta}$	(Eff, Tox)	BOP2	TSE	BOP2	TSE	BOP2	TSE
1	(0.15, 0.30, 0.15, 0.40)[§]	(0.45, 0.30)	9.7	9.6	85.4	89.3	27.1	19.0
2	(0.20, 0.30, 0.15, 0.35)	(0.50, 0.35)	6.3	7.0	89.4	91.9	25.1	17.8
3	(0.10, 0.30, 0.15, 0.45)	(0.40, 0.25)	7.1	7.6	88.9	91.5	25.0	18.0
4	(0.15, 0.35, 0.10, 0.40)	(0.50, 0.25)	36.7	29.4	58.1	69.9	38.6	28.2
5	(0.15, 0.35, 0.05, 0.45)	(0.50, 0.20)	49.6	38.3	46.2	61.3	42.9	32.0
6	(0.15, 0.40, 0.05, 0.40)	(0.55, 0.20)	69.6	54.2	28.3	45.6	49.2	38.8
7	(0.18, 0.42, 0.02, 0.38)[†]	(0.60, 0.20)	80.2	64.4	18.4	35.4	52.8	43.4
8	(0.15, 0.50, 0.05, 0.30)	(0.65, 0.20)	85.3	73.7	13.5	26.2	54.8	47.8

PRN, percentage of rejecting the null; PET, percentage of early termination; BOP2, Bayesian optimal phase II; Eff, efficacy; Tox, toxicity.
[§] null hypothesis.
[†] alternative hypothesis.

Table DII. Stopping boundaries of the BOP2 design with jointly monitoring efficacy and toxicity endpoints as described in trial example 4.

		Number of patients treated										
	Stop the trial if	10	15	20	25	30	35	40	45	50	55	60
or	# of Responses ⩽	3	5	7	10	12	15	17	20	22	25	27
	# of Toxicities ⩾	5	6	8	9	10	12	13	14	16	17	18

Maximum sample size is 60, and cohort size is 5.
BOP2, Bayesian optimal phase II.

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